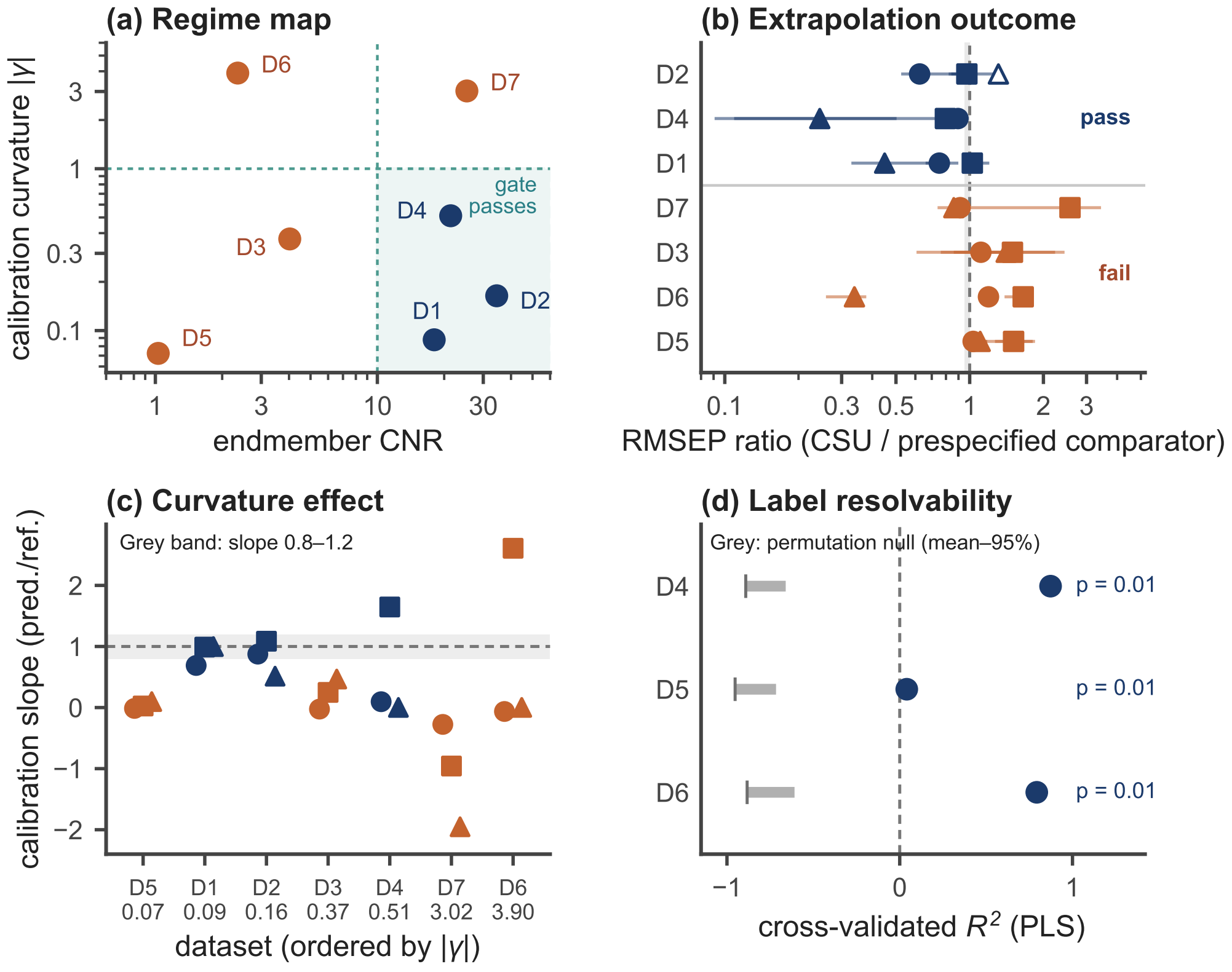
(a) Regime map
calibration curvature |γ|
3
1
0.3
0.1
D6
D7
D4
D3
D2
D1
D5
gate
passes
1
3
10
30
endmember CNR
(b) Extrapolation outcome
D2
D4
D1
D7
D3
D6
D5
pass
fail
0.1
0.3
0.5
1
2
3
RMSEP ratio (CSU / prespecified comparator)
(c) Curvature effect
Grey band: slope 0.8–1.2
calibration slope (pred./ref.)
2
1
0
−1
−2
D5 0.07
D1 0.09
D2 0.16
D3 0.37
D4 0.51
D7 3.02
D6 3.90
dataset (ordered by |γ|)
(d) Label resolvability
Grey: permutation null (mean–95%)
D4
D5
D6
p = 0.01
p = 0.01
p = 0.01
−1
0
1
cross-validated R^2 (PLS)